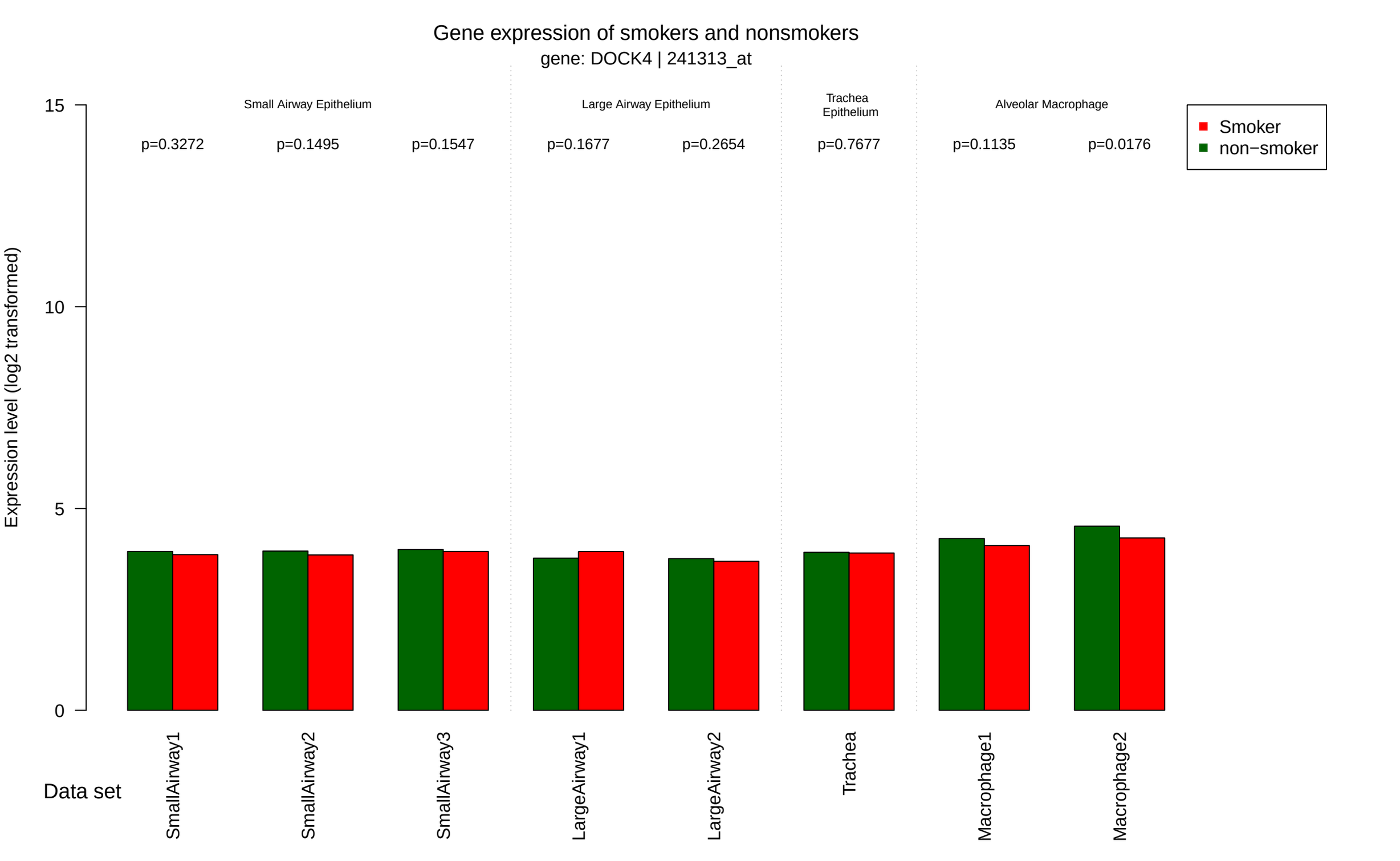
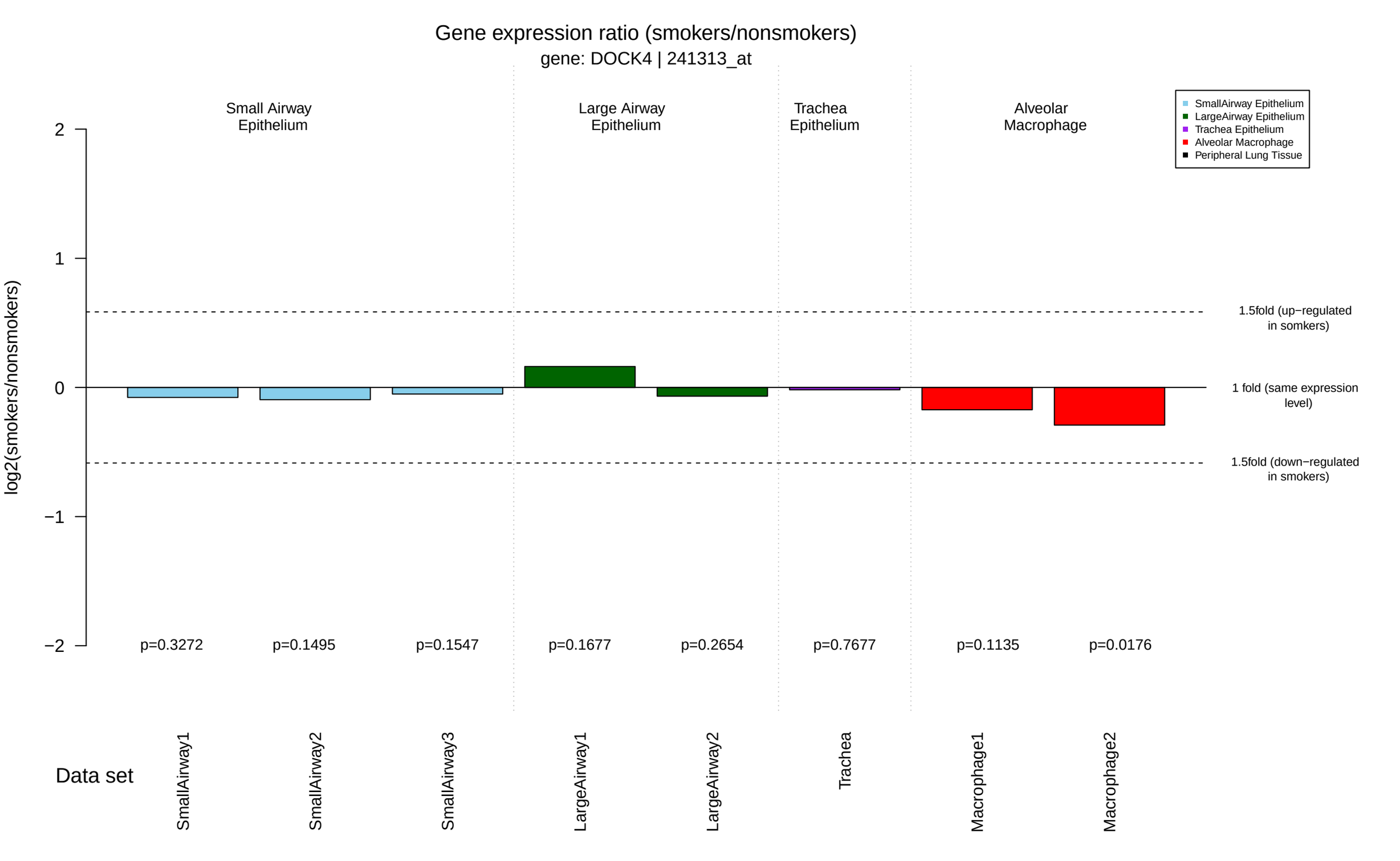
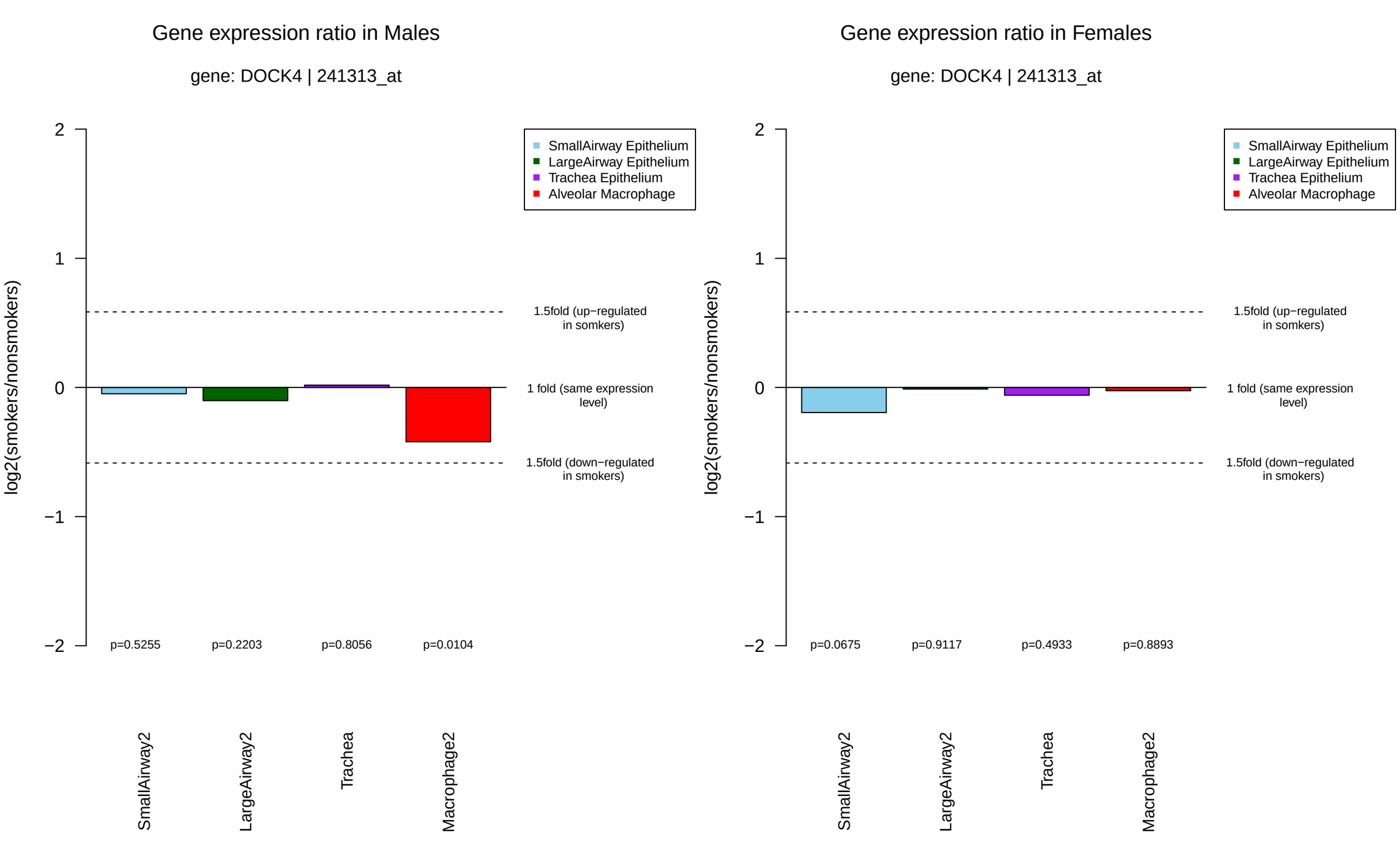
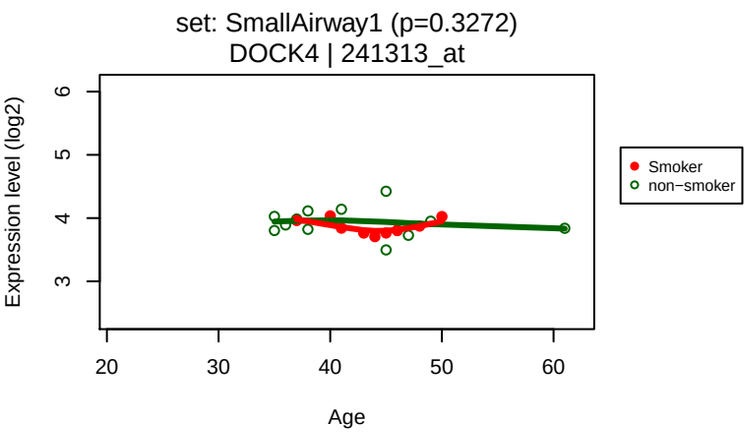
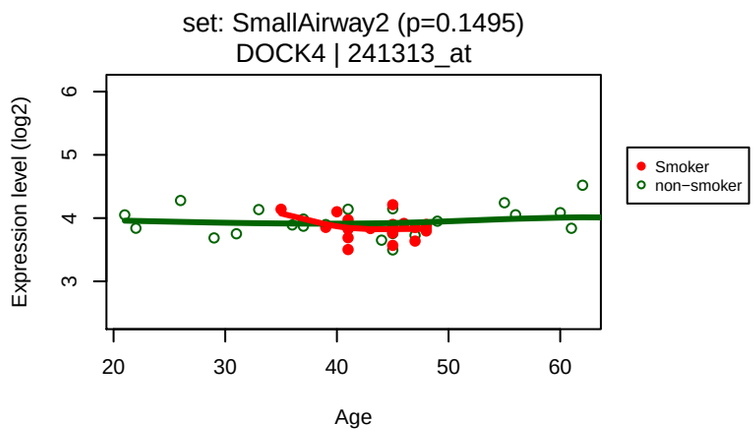
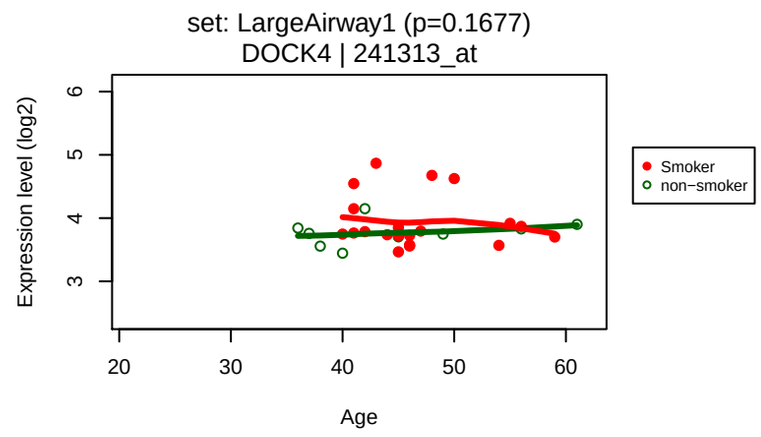

Gene expression of smokers and nonsmokers
gene: DOCK4 | 241313_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3272
p=0.1495
p=0.1547
p=0.1677
p=0.2654
p=0.7677
p=0.1135
p=0.0176
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

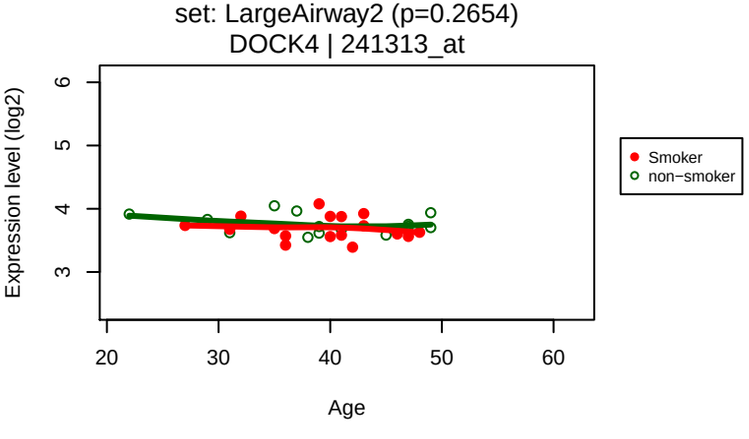

Gene expression ratio (smokers/nonsmokers)
gene: DOCK4 | 241313_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3272
p=0.1495
p=0.1547
p=0.1677
p=0.2654
p=0.7677
p=0.1135
p=0.0176
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

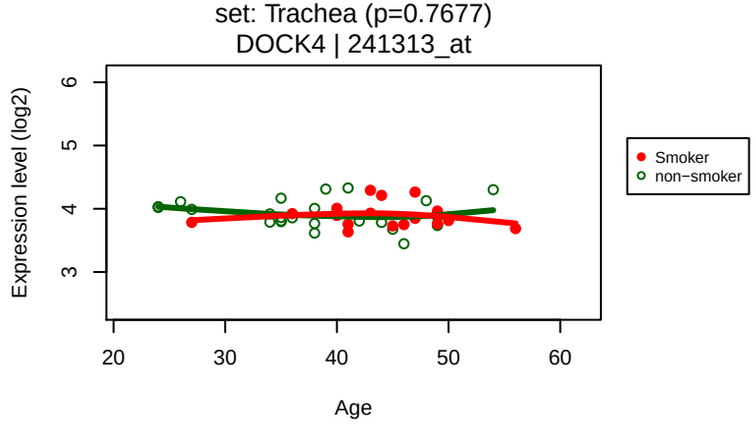

Gene expression ratio in Males
gene: DOCK4 | 241313_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5255
p=0.2203
p=0.8056
p=0.0104
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DOCK4 | 241313_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0675
p=0.9117
p=0.4933
p=0.8893
SmallAirway2
LargeAirway2
Trachea
Macrophage2

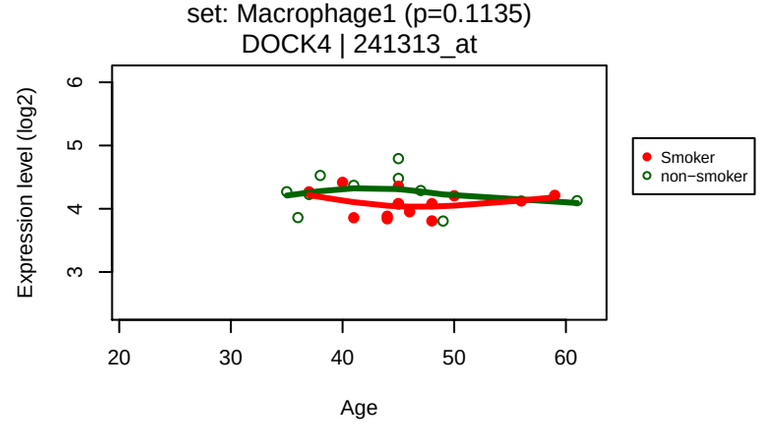

set: SmallAirway1 (p=0.3272)
DOCK4 | 241313_at
Expression level (log2)
Age
Smoker
non−smoker

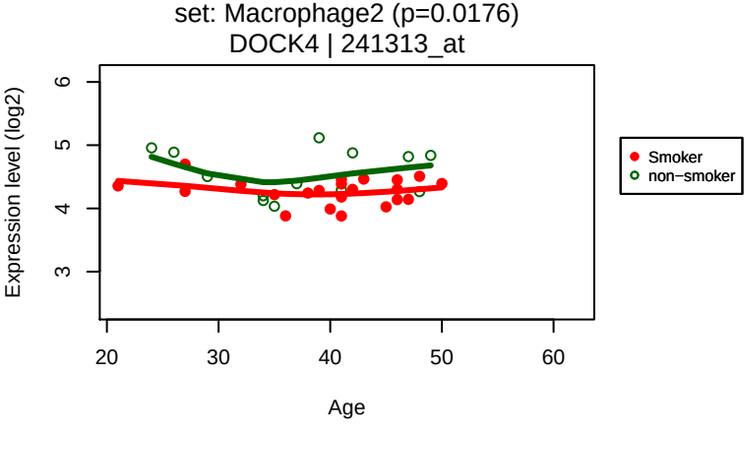

set: SmallAirway2 (p=0.1495)
DOCK4 | 241313_at
Expression level (log2)
Age
Smoker
non−smoker

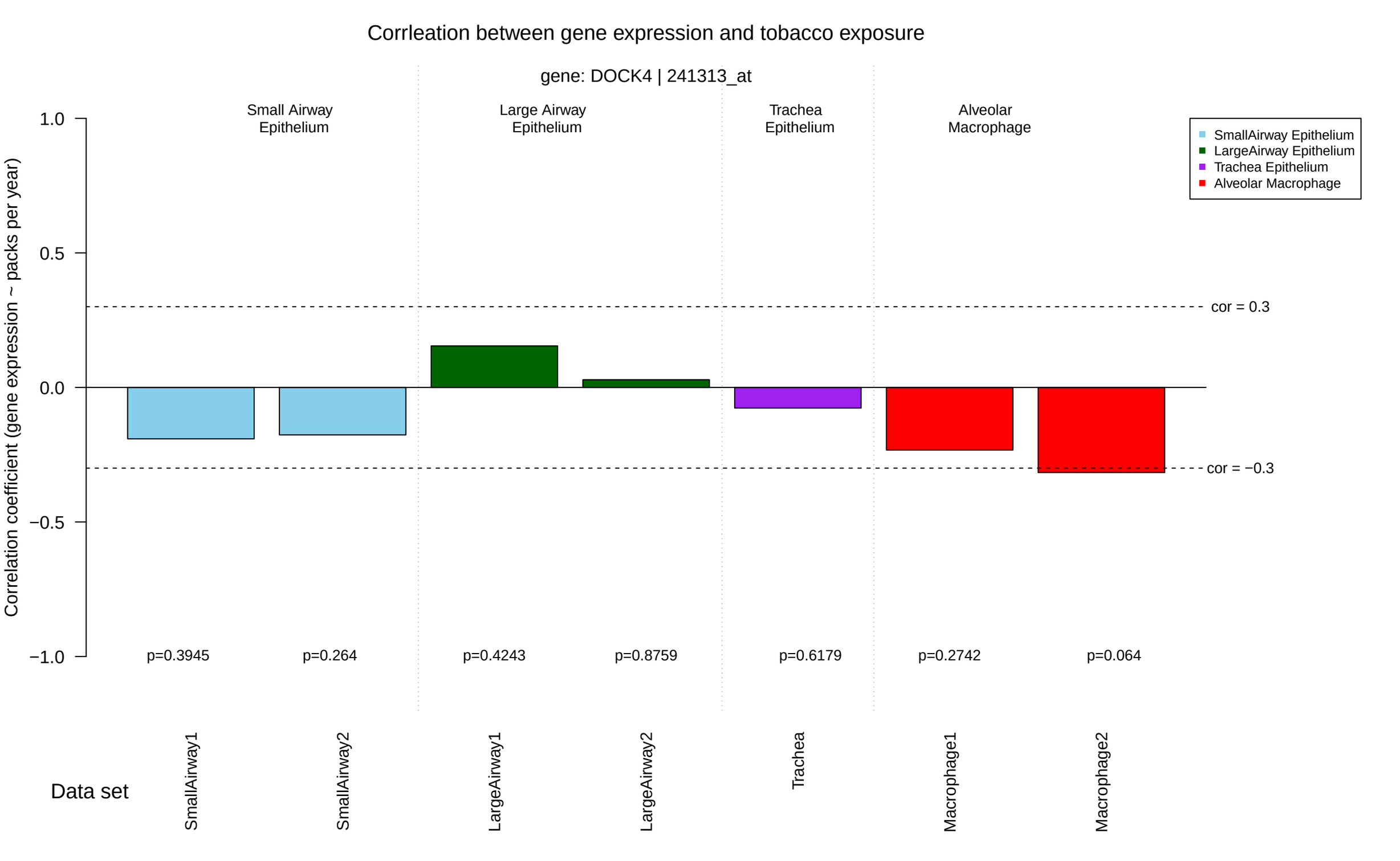

set: LargeAirway1 (p=0.1677)
DOCK4 | 241313_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2654)
DOCK4 | 241313_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7677)
DOCK4 | 241313_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1135)
DOCK4 | 241313_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0176)
DOCK4 | 241313_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DOCK4 | 241313_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3945
p=0.264
p=0.4243
p=0.8759
p=0.6179
p=0.2742
p=0.064
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2